MCO — METACHRONAL CASCADE ORNITHOPTER

Master Design Specification · Issue 1.0 · 2026-06-03

Consolidated specification covering MCO-1D drone prototype and MCO-1A full-scale piloted SAR aircraft.

1. Programme Overview

The MCO programme develops a forward-flight ornithopter using an 8-wing metachronal cascade. The aircraft flies like a large bird — not a drone. Hover is not a design requirement. The MCO-1D is a 3.45 kg drone that validates every aerodynamic and control mode before the MCO-1A crewed SAR aircraft is built. The MCO-1D is the mandatory flight test programme for the MCO-1A.

MCO-1A Piloted SAR aircraft · 376 kg MTOW (hybrid) · 924mm chord · FX 63-137 + Gurney · CBR600RR series hybrid Programme ru

2. MCO-1D — Locked Drone Specification (Rev G, frozen)

2.1 Wing geometry

Parameter	W1	W2	W3	W4	Units
Semi-span	772	749	735	712	mm
Tip-to-tip	1,544	1,498	1,470	1,424	mm
Chord (all)	165	165	165	165	mm
Aspect ratio	4.7	4.5	4.5	4.3	—
Airfoil (all)	S1223	S1223	S1223	S1223	UIUC exact
VGs	17%c	17%c	17%c	17%c	h=8.4mm, 18° toe, pairs
Gurney flap	1%c	1%c	1%c	1%c	1.65mm, integral rib tail
Raked wingtip	25°/15%c	25°/15%c	25°/15%c	25°/15%c	18mm extension

Total wing area: 0.979 m² · Mean AR: 4.5 · CL,max (VGs+Gurney): 2.40

2.2 Drivetrain

Component	Specification	Qty	Notes
Flap motor	GoolRC 5010-360KV	8	90g each, 6S direct
Motor controller	ODrive Micro	8	CAN-FD bus, 6S 22.2V
Reducer	7:1 single stage	8	12mm drive → 84mm driven, 1.8mm Dyneema
Sweep servo	ZOSKAY DS3225 25KG	8	60g, 2.40 N·m, 270° range
Pitch servo	Emax ES08MD II	8	8.5g, digital, 0.08s/60°
Wave library CPU	Raspberry Pi Zero 2W	1	Python wave library
Safety FCC	Pixhawk Mini	1	MAVLink, pitot, stall protect
Battery	6S LiPo 5000mAh	1	~490g, ~20 min cruise

2.3 Mass budget

Item	Mass g	Item	Mass g
Motors ×8	720	Wings ×8	~726
ODrive Micro ×8	80	Fuselage	200
Reducers ×8	224	FCC + GPS	60
ZOSKAY ×8	480	Wiring	30
Emax ×8	68	Battery	490
Root hardware ×8	200		

Item	Mass g	Item	Mass g
Rope ×8	**80**	**MTOW**	**3,448**

Wing loading: 0.73 lb/ft² · Stall: 4.8 m/s · Cruise: 6–7 m/s at 0.8 Hz · Sprint: 12–15 m/s at 3 Hz · Stoop (STOOP-2A): ~36 m/s

3. MCO-1A — Piloted SAR Aircraft Specification

3.1 Wing geometry

Parameter	W1	W2	W3	W4	Units
Semi-span	4,324	4,199	4,116	3,991	mm
Tip-to-tip	8,648	8,398	8,232	7,982	mm
Chord (all)	924	924	924	924	mm
Aspect ratio	4.7	4.5	4.5	4.3	—
Airfoil (all)	FX 63-137	FX 63-137	FX 63-137	FX 63-137	Wortmann
VGs	None	None	None	None	Not needed at Re >5×10■
Gurney flap	1%c	1%c	1%c	1%c	9.24mm, CFRP TE doubler
Raked wingtip	25°/15%c	25°/15%c	25°/15%c	25°/15%c	139mm ext. — Re review req.
Stow angle	210°	210°	210°	210°	Hard stop + spring fairing

Parameter	Value	Notes
Total wing area	30.7 m² (331 ft²)	8 wings, uniform 924mm chord
Wing loading	2.0 lb/ft² (98 N/m²)	Peregrine-class — speed + loiter balance
CL,max (FX+Gurney)	~2.07	At Re 5×10■; Gurney adds ~0.10 to clean 1.97
Best L/D	~16:1	FX 63-137 at Re 10■; vs ~12:1 S1223
Stall speed	~8.7 m/s (17 kt)	All wings deployed, MTOW 376kg
Scale factor from 1D	7.5×	165mm → 924mm chord

3.2 Powertrain — CBR600RR Series Hybrid

The MCO-1A uses a series hybrid architecture: the IC engine drives a generator only and never connects mechanically to the thrust system. All thrust comes from the 8 wing-flap motors, unchanged from the MCO-1D architecture. The generator charges a sprint battery and supplies the motor bus continuously.

Component	Specification	Mass	Notes
IC engine	Honda CBR600RR stripped	~50 kg	119 hp / 89 kW peak; 70 kW at 60% continuous cruise
Generator	3-phase PMG, coupled direct	~8 kg	No PSRU needed — generator runs at engine RPM (9,000 rpm)
Exhaust + mount	Stainless + CFRP	~4 kg	Routed to fuselage underside
Sprint battery	Li-ion 5 kWh	~12 kg	200 Wh/kg cells; 6 min full-sprint reserve; recharges in 5 min at loiter
Fuel	MOGAS 91, 15L	~11 kg	~45-60 min cruise endurance
Wing motors (×8)	Large brushless, crank-and-rod	—	Same architecture as MCO-1D, scaled up
FCC	RPi Zero 2W + Pixhawk Mini	—	Identical wave library software to MCO-1D

Total powertrain system mass: ~85 kg · vs H2 fuel cell (9 kg) → MTOW rises from 300 kg to 376 kg · Cost: ~$5,000–6,000 total (vs $27,000 for Rotax 912 hybrid)

3.3 Fuselage

Parameter	Specification
Form	Enclosed teardrop monocoque, CFRP
Cockpit	Pilot reclines 30°, polycarbonate canopy, 5-point harness
CD × S (fuselage)	0.12 m² (Pipistrel Virus SW class)
Wing root fairings	Spring-loaded CFRP clamshell, closes at 210° stow

Parameter	Specification
Fairing mass	~420 g per station × 8 = ~3.4 kg total
Length (est.)	~2.5 m
Fuselage lifting body	~10–15% lift contribution at sprint speeds

3.4 Performance envelope

Mode	Speed	Power	Notes
Stall (all wings)	8.7 m/s 17 kt	—	CL,max 2.07, MTOW 376 kg
Best endurance (full wings)	~12 m/s 23 kt	~5 kW	Max loiter, SAR search
Best range (full wings)	~16 m/s 31 kt	~8 kW	Normal transit
W3+W4 cruise (partial stow)	24 m/s 46 kt	~18 kW	W1+W2 faired
GA sprint — generator (135°)	70 m/s 136 kt	70 kW	Cessna 172 class ✓
GA sprint — + battery burst	82 m/s 159 kt	~100 kW	Piper Arrow class ✓
STOOP-2A (all faired, ballistic)	~80 m/s 156 kt*	0	Aerodynamic terminal
Best glide (FX+Gurney)	~19 m/s 37 kt	0	L/D ~16:1
Altitude cruise (4,000m)	30 m/s IAS → 37 m/s TAS	~10 kW	22% more range/litre

* 156 kt stoop is aerodynamic terminal velocity with enclosed cockpit + faired wings. Structural and pilot load limits govern operational ceiling.

4. Wave Library — Confirmed Mode Catalogue

The wave library is identical software on both MCO-1D and MCO-1A. All parameters below apply to MCO-1A; MCO-1D values are 7.5× higher in frequency and use 165mm chord geometry. Addenda Rev A (Tweaks 3+4) and Rev B (Tweak 2) are incorporated.

4.1 Speed control — variable sweep schedule (Addendum Rev A+B)

Speed lever	Airspeed	W1	W2	W3	W4	Mode
Glide detent / 0–30%	<32 kt	60°	60°	60°	60°	Full deploy — soar / slow cruise
30–50%	32–46 kt	→120°	60°	60°	60°	W1 sweeping aft
50–65%	46–68 kt	210°†	210°†	60°	60°	W3+W4 cruise — swift mode
65–80%	68–97 kt	210°†	210°†	→135°	→135°	All sweeping to delta
80–95%	97–120 kt	210°†	210°†	135°	135°	GA sprint
95–100%	>120 kt	210°†	210°†	→165°	→165°	Pre-stoop
STOOP switch	—	210°†	210°†	210°†	210°†	Full stow — STOOP-2A

† 210° with spring-loaded root fairing closed (Tweak 2). FCC gates 68+ kt modes on FAIRINGS_FITTED = True.

4.2 Dihedral bias at high sweep (Tweak 4 — Addendum Rev A)

Sweep angle	Dihedral bias	Flap amplitude	Notes
60°–120°	0°	±80°	Normal ornithopter — no bias
120°–130°	–4°	±40°	Bias onset
130°–140°	–8°	±20°	Swift-mode oscillation begins
140°–150°	–12°	±10°	Optimal thrust plane at 100–120 kt
150°–165°	–15°	±5°	Pre-stoop minimal motion
165°–210°	Stoop pitch DoF	Active	STOOP-2A active teardrop

4.3 Stoop dive sequence

Mode	Wings	Speed range	Purpose
STOOP-1	Progressive 60°→210°, W4→W1, 0.2s offsets, 60°/s	0→dive	Wing fold; FCC pitches nose down
STOOP-2A	All 210°, pitch/sweep DoF active	→156 kt terminal	Active teardrop, max speed

Mode	Wings	Speed range	Purpose
STOOP-2B	W1+W2 at 210°, W3+W4 at 135° cupped	80→240 km/h	Full authority controlled descent
STOOP-4A	W3+W4 deploy 90°, 0.3 Hz LEV flap	240→80 km/h	M-shape LEV pullout, 3g available
STOOP-4B	All wings deploy, 1.0 Hz	80→20 km/h	Flare and decelerate
STOOP-4C	All wings 90°→60°	<80 km/h	Return to cruise

Stow order W4→W3→W2→W1 is LOCKED in FCC firmware — cannot be reversed without wing collision at 138mm root stagger. FCC hard-coded STOOP-4A trigger at 227m AGL if pilot has not initiated pullout.

5. Key Design Decisions (Locked)

Decision	Rationale	Status
No hover requirement	At hover demand 8 motors draw 35A = 5,328W — no practical battery. In forward flight at 6.5 m/s motors draw ~2A = ~200W. The	LOCKED
7:1 single-stage reducer on MCO-1D	At 2A cruise draw the 7:1 is thermally fine. The 22:1–30:1 detour was caused by incorrectly using 35A hover current. Original wa	LOCKED
165mm chord on MCO-1D	Derived from confirmed drivetrain mass of 2,608g. Chord history: 120→260→165mm. 165mm is correct.	LOCKED
S1223 + VGs + Gurney on MCO-1D	S1223 optimised for Re 2×10■. VGs energise BL at 17%c. Gurney adds 0.10 CL,max. Confirmed CL,max = 2.40.	LOCKED
FX 63-137 + Gurney on ALL MCO-1A wings	MCO-1A operates at Re 5×10■–4×10■. FX 63-137 has equal or better CL,max AND 10% lower profile drag above Re 5×10■. C	LOCKED
CBR600RR series hybrid for MCO-1A	119 hp / 89 kW at 50 kg. Drives generator direct at 9,000 rpm (no PSRU needed in series hybrid architecture). Cost ~$2,000–5,000 vs $	LOCKED
Hard stop at 210° (wing stow)	Aerodynamic load at full stow tries to drive wing from 210° to 180°. Stop resists this. Does not interfere with control inputs (which	LOCKED
No tip retention clip	Clip solved galloping for passive teardrop (eliminated mode). Active servo at 210° provides 3.7 Hz effective natural frequency —	LOCKED
W1+W2 span extension (1.30×) on MCO-1A	Better loiter L/D (16:1 vs 13:1), lower stall (16 vs 18 kt), no sprint speed penalty (wings stowed). Defer to MCO-1D flight validation be	PENDING

6. MCO-1D Validation Test Plan

Every mode must be validated on the MCO-1D before the MCO-1A is built. The MCO-1D is not a demonstrator — it is the MCO-1A's mandatory flight test programme. The intelligence transfers completely; the risk does not.

Test ID	Mode	Success criterion	Gates MCO-1A item
T1	Metachronal 90° cruise (0.8 Hz)	Stable forward flight >30 sec at 6–7 m/s	Cascade phase offset validation
T2	Speed lever sweep schedule (Tweak 3)	Airspeed increases smoothly with lever	Variable sweep control law
T3	W3+W4 partial stow cruise	Sustains 35 kt level flight for 30 sec	W3+W4 sprint mode for MCO-1A
T4	135° swept oscillation (Tweak 4)	Airspeed increases without altitude loss	Confirms swept thrust mechanism
T4-CT	CT measurement at 135° sweep	Fit CT from T = CT·q·S at measured speed	PRIMARY input: MCO-1A 120 kt claim
T5	Stoop-1 wing fold sequence	All 8 wings reach 210° in correct W4→W1 order	Stoop sequence timing and servo torque
T6	Stoop-2A + pullout (STOOP-4A/4B/4C)	Full stoop and recovery without structural event	Full stoop sequence MCO-1A
T7	Asymmetric sweep turns	Coordinated banked turns L and R	Turn authority model for MCO-1A
T8	Glide detent	Stable glide >9:1 at 8 m/s	Glide ratio baseline before MCO-1A
T9	STOOP-4A galloping check	No stowed-wing oscillation at any speed	Active servo galloping suppression

Test T4-CT is the single highest priority test — it determines whether the MCO-1A achieves 120 kt in level flight or requires a gentle descent. All other performance claims are validated by known aerodynamic relationships.

7. Open Items

#	Item	Priority	Dependency
O-1	MCO-1D hard stop reposition: 200°→210° (one workshop op)	High	Before any sprint/stoop test
O-2	wave_library.py: SWEEP_MAX_DEGREES=210, STOW_ANGLE=210	High	Software, 2 lines
O-3	MCO-1D flight test programme (T1–T9 above)	Critical	After O-1, O-2
O-4	MCO-1A: Galloping re-analysis for extended W1+W2 (if pursued)	High	Before W1+W2 extension commit
O-5	MCO-1A: Stow servo torque verification for extended W1+W2	High	Before W1+W2 extension commit
O-6	MCO-1A: Root CFRP layup uprate (+69% bending moment, ext. W1+W2)	High	Structural design phase

#	Item	Priority	Dependency
O-7	MCO-1A: Raked tip geometry review at Re 10■ (vs Re 10■ drone data)	Medium	Before MCO-1A wing design freeze
O-8	MCO-1A: CBR600RR engine procurement and generator matching	Medium	After MCO-1D flight validation
O-9	MCO-1A: FAA Experimental Airworthiness Certificate application	Medium	After MCO-1D validation complete
O-10	MCO-1A: CFRP wing construction specification (replaces LW-PLA)	Medium	MCO-1A structural design phase

8. Supporting Document Register

Document	Issue	Status	Contents
MCO_wave_parameter_library.pdf	Base	Released	Full wave library mode catalogue, original Rev G
MCO_wave_library_addendum.pdf	Rev A	Released	Tweaks 3+4: variable sweep schedule, dihedral bias
MCO_wave_library_addendum_revB.pdf	Rev B	Released	Tweak 2: 210° stow extension, spring-loaded fairings
MCO_1D_Wing_Construction.pdf	Rev G	Released	Wing build specification, S1223, VGs, Gurney, raked tip
MCO_1D_Rib_Drawing.pdf	Rev G	Released	Rib geometry and construction details
MCO_1D_VG_Design.pdf	Rev E	Released	VG placement, geometry, installation method
MCO_Master_Specification.pdf	**Issue 1.0**	**THIS DOCUMENT**	**Consolidated programme specification**